

SENTIMENT ANALYSIS ON HOTEL REVIEWS

*A project report submitted to ICT Academy of
Kerala in partial fulfillment of the
requirements
for the certification of*

CERTIFIED SPECIALIST IN DATA SCIENCE & ANALYTICS

Submitted by

ALOK KRISHNAN



ICT ACADEMY OF KERALA
THIRUVANANTHAPURAM, KERALA, INDIA
Sept 2024

List of Figures

Index	Heading	Page number
1	Distribution of Ratings	8
2	Proportion of Reviews by Continent	9
3	Top 20 Nationalities	10
4	Distribution of ratings in UK	11
5	Sentiment Distribution	12
6	Confusion Matrix Heatmap	14

List of Abbreviations

1. **NLP**: Natural Language Processing
2. **NLTK**: Natural Language Toolkit
3. **TF-IDF**: Term Frequency-Inverse Document Frequency
4. **SMOTE**: Synthetic Minority Over-sampling Technique
5. **SVC**: Support Vector Classification
6. **XGB**: eXtreme Gradient Boosting

Table of Contents

Index	Heading	Page number
0	Abstract	5
1	Problem Definition	6
2	Introduction	7
3	Literature Survey	16
4	Result	18
5	Conclusion	19
6	Future Work	19
7	References	20

ABSTRACT

This project focuses on performing sentiment analysis on hotel reviews within the travel and tourism sector. Utilizing Python and libraries such as `pandas`, `seaborn`, and maybe `Streamlit`, the project classifies reviews into positive, negative, or neutral sentiments. The project conducts a thorough analysis of customer feedback to classify reviews into positive, negative, or neutral sentiments. The analysis includes examining sentiment distributions, identifying common themes, and summarizing key insights from the data. The primary goal is to provide a detailed understanding of guest sentiments, uncovering trends and patterns that can inform hotel management decisions and enhance the quality of service, possibly using an interactive web application. By providing a comprehensive view of customer feedback, the application aims to assist hotel management in understanding guest experiences and identifying key areas for improvement, ultimately enhancing decision-making and service quality in the hospitality industry.

1. Problem Definition

1.1 Overview

In the competitive landscape of the travel and tourism industry, understanding and improving customer satisfaction is crucial for maintaining a strong market position. Hotels and other hospitality businesses often rely on guest reviews to gauge their service quality and identify areas for improvement. However, the sheer volume of feedback makes it challenging to systematically analyze and extract actionable insights from these reviews. Traditional methods of review analysis are time-consuming and may not fully capture the nuanced sentiments expressed by guests. This project seeks to address these challenges by utilizing sentiment analysis to provide a comprehensive understanding of customer feedback by classifying reviews into positive, negative, or neutral sentiments.

1.2 Problem Statement

The problem this project aims to solve is the need for an automated, scalable solution to classify customer reviews into distinct sentiment categories—positive, neutral, and negative. This classification will enable businesses to quickly and accurately gauge customer sentiment, identify patterns in feedback, and take proactive measures to address concerns or capitalize on positive trends. By implementing sentiment analysis, businesses can improve service quality, enhance customer experiences, and make data-driven decisions that are essential for maintaining a competitive edge in the industry.

This sentiment analysis project not only provides a solution to the problem of data overload but also empowers businesses to act on insights derived from customer feedback, ultimately leading to better service, higher customer satisfaction, and stronger brand loyalty. Once the classifier is built, a hotel can use it to analyze data from their own database, classify based on sentiments, and derive insights to make their service better.

2. Introduction

The primary objective of the project is to develop a machine learning model that accurately classifies the sentiment of customer reviews into three categories: positive, neutral, or negative. This classification will serve as a basis for later understanding overall customer satisfaction and areas of improvement.

The secondary objective is to analyze the distribution of sentiments across the dataset to gain insights into the general sentiment trends. This analysis will help in understanding the proportion of positive, neutral, and negative feedback within the data and then build and deploy a very basic user-friendly web application using Streamlit, allowing users to input new reviews and receive instant sentiment analysis results. This tool will make sentiment analysis accessible to business stakeholders because they can tailor the application to suit the needs of their own database.

2.1 Dataset Description

The dataset is taken from Kaggle datasets. The Booking.com dataset is a collection of hotel ratings and reviews collected from Booking.com. The dataset contains various columns that provide information about each review, including review title, reviewed by (name/username of the reviewer), tags or labels associated with the review, average rating of the hotel, number of images attached to the review, url of the review page, text content of the review, nationality of the reviewer, crawled date and time (when reviewed data was obtained), hotel name and its url. The dataset can be used for sentiment analysis to analyze the overall sentiment and opinions of hotel guests. By analyzing the review text and ratings, one can determine whether guests had a positive or negative experience and this is what we use to build a classifier.

[Dataset link](#)

The dataset contains 15 columns and 26675 rows. For the purposes of building our classification model, we have used mainly 3 columns:

- review_title: The title given by a reviewer for their review. (Text)
- review_text: The text content of each review without any processing or modifications. (Text)
- Rating: the rating given by individual reviewer in a review

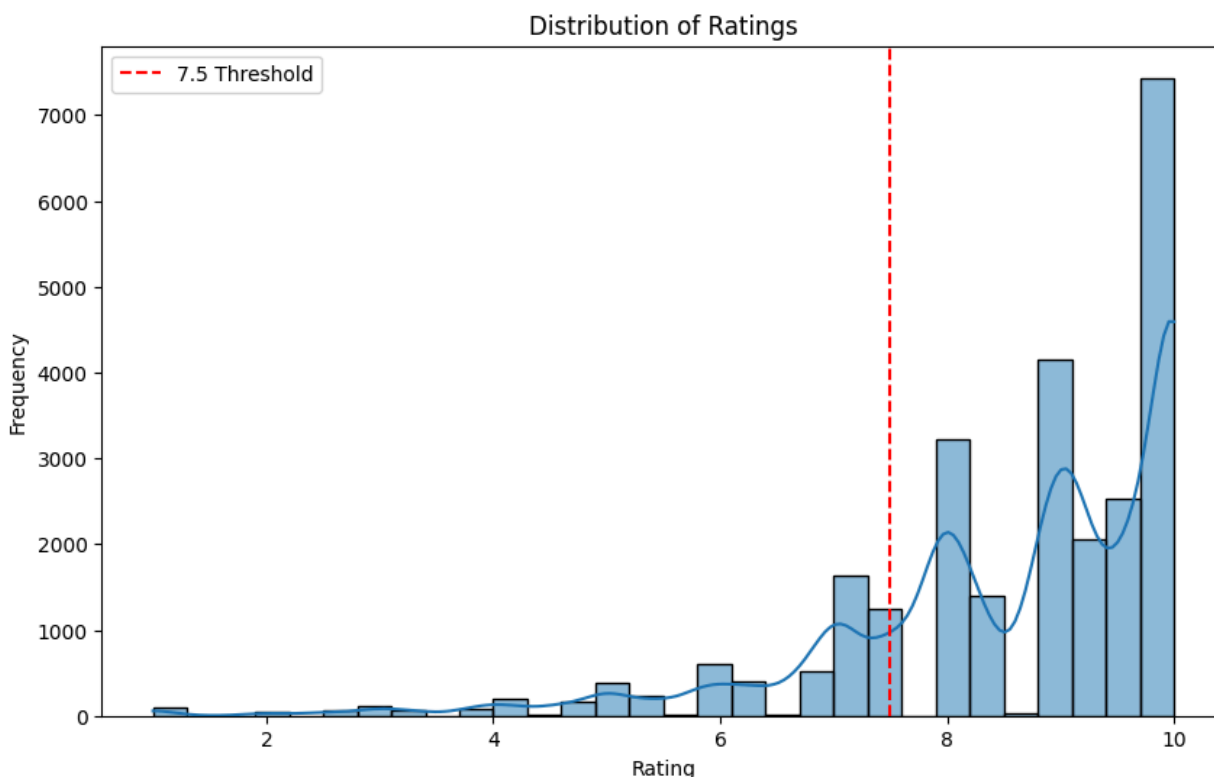
2.2 Data Preprocessing, Cleaning and Exploration

The focus was primarily placed on the three key columns mentioned earlier. Additionally, the *nationality* column, which indicates the reviewer's nationality, was utilized to derive further insights. The dataset contained several missing values. For the text-based columns, null values were replaced with the placeholder text "No review provided."

It was observed that after grouping by review text, there were already 7,687 rows in the review text column containing the phrase "There are no comments available for this review." As these rows were more likely to introduce noise rather than valuable content for the classifier, they were removed along with other rows containing null values.

For the null values present in the rating column, these were replaced with the median value. This decision was based on the observation of a skewed distribution in the histogram plot of the ratings.

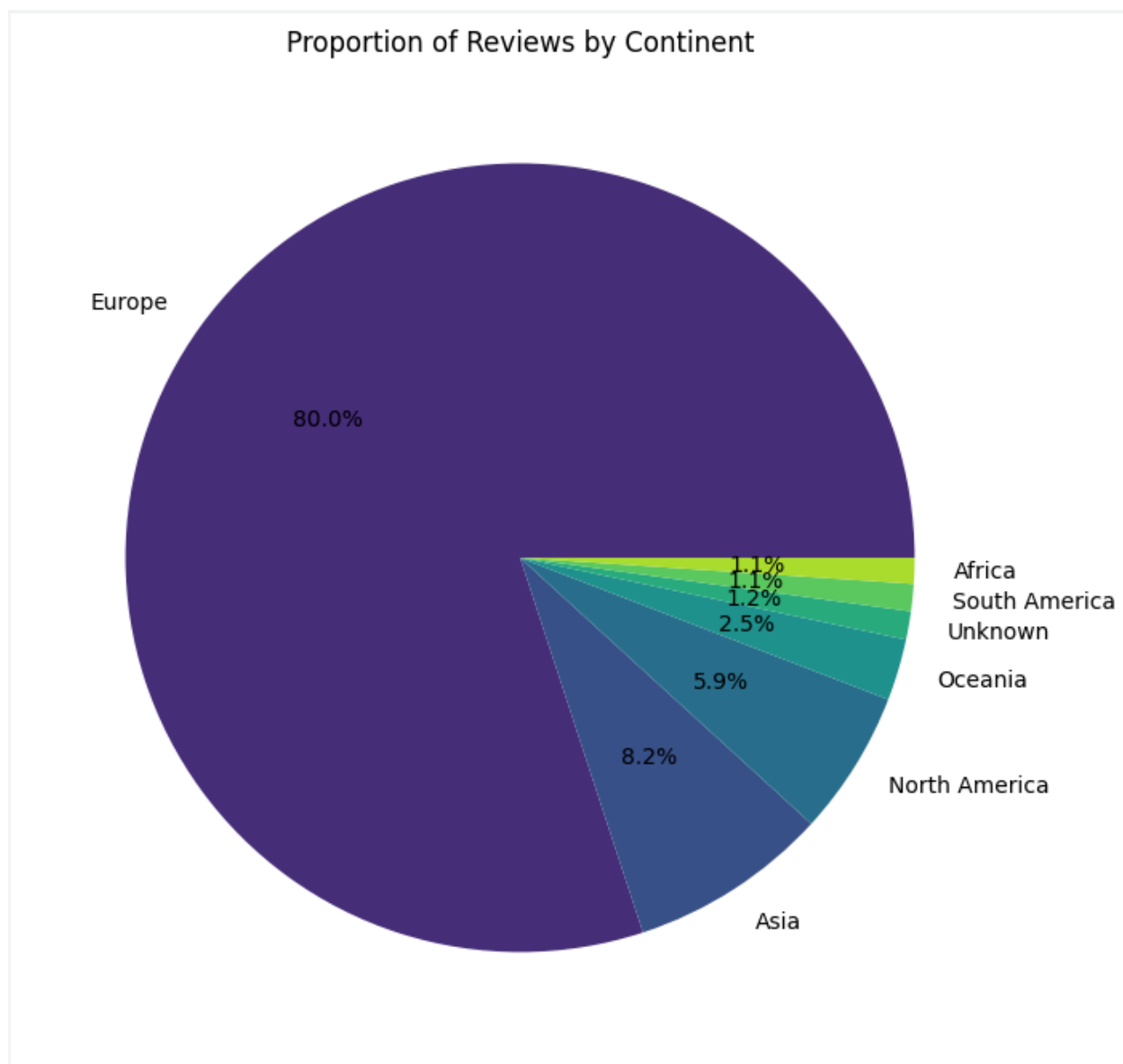
The rating values were clipped to the range of 1 to 10.



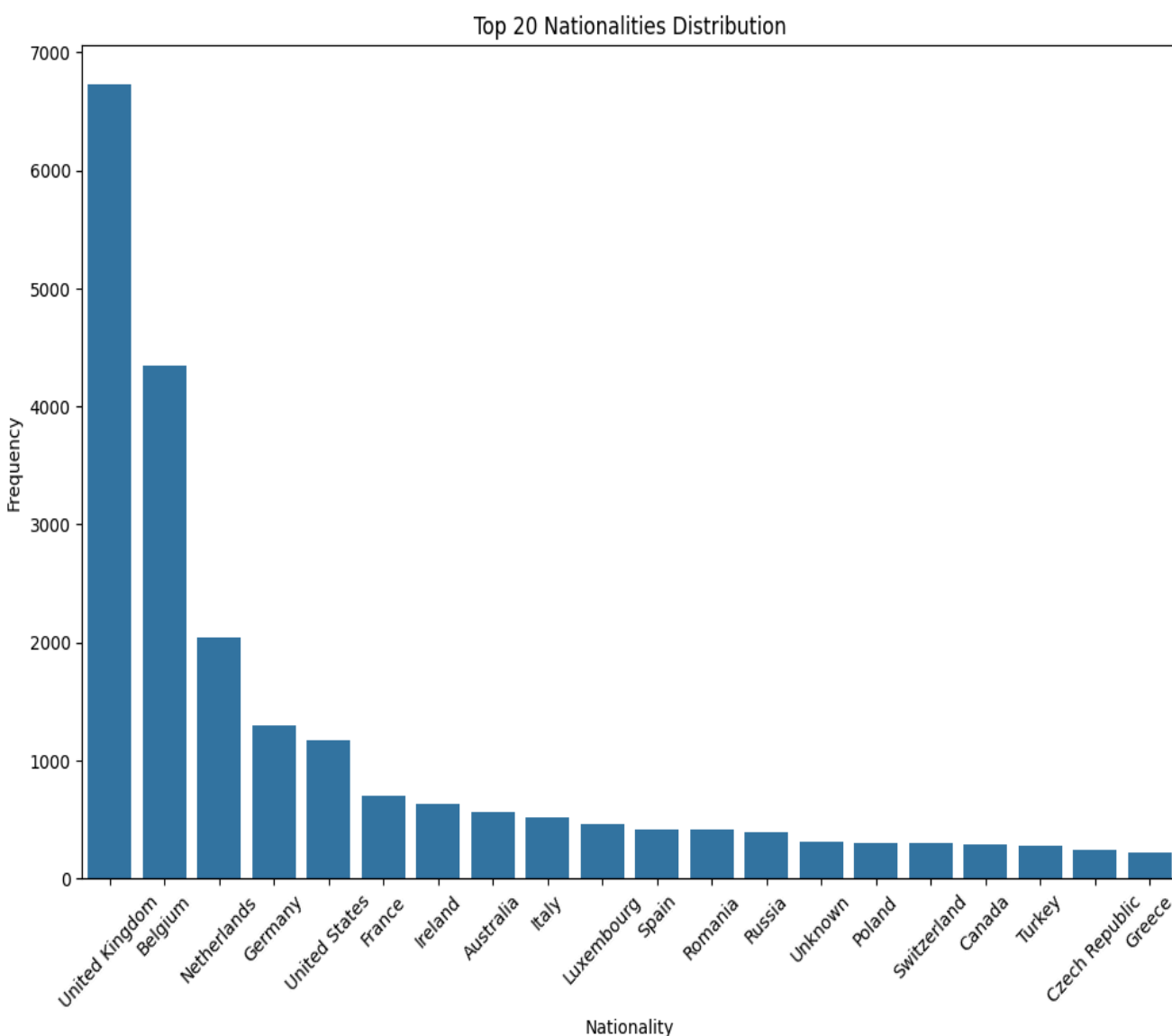
After addressing the issues mentioned above, a check for duplicates was conducted, and none were found. Subsequently, insights were derived by plotting data using the *nationality* column.

It was found that there were reviewers from 160+ countries in the dataset. A dictionary was made for all the countries from the *nationality* column. after separating unknown values, they were used to create a column named *continent* and added it to the dataset.

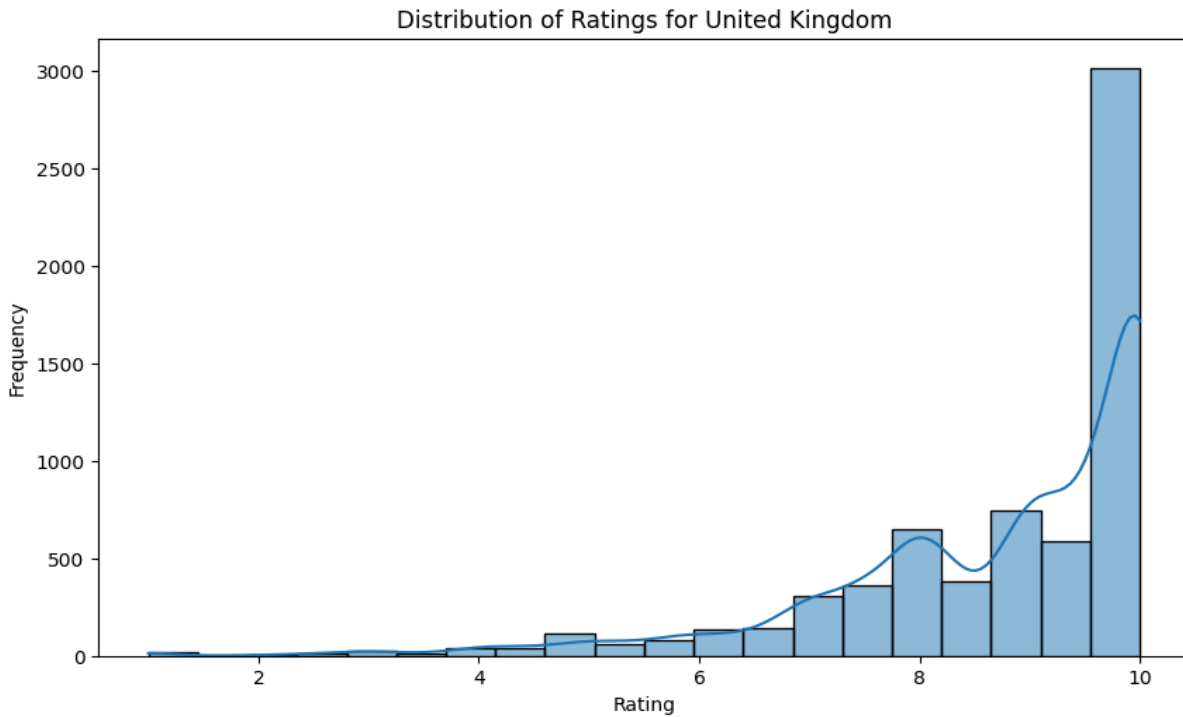
On plotting a pie chart, it was discovered that the majority of the reviewers were European.



A plot was done to find the top 20 countries which had the most reviewers. It was found that the majority of the reviewers were from the United kingdom, so special attention may be needed for their reviews if they are the most frequent customers to check if they are satisfied.



A plot was done to check how the reviewers from the U.K generally rate the hotels and the results found that most of them are positive, so it can be understood that if a customer from the Uk does give a negative review, attention is required.



2.2 Text Processing and Feature Engineering

Natural Language Processing (NLP) is essential for leveraging the textual information to provide input data for the sentiment analysis model. For the *review_title* and *review_text* columns, several text preprocessing steps were applied, including tokenization, lowercase conversion, whitespace removal, stopword removal, and lemmatization using the NLTK library. These two columns were then combined into a single *combined_review* column to enhance feature extraction.

The primary goal of the project is to build a sentiment classifier. To achieve this, a sentiment label column was created to represent positive, negative, and neutral sentiments, which were derived from the rating column. Specifically:

- Ratings below 5 were labeled as 0, representing negative sentiment.
- Ratings between 5 and 7 were labeled as 1, representing neutral sentiment.
- Ratings above 7 were labeled as 2, representing positive sentiment.

With the two key columns (*combined_review* and *sentiment*) prepared, the focus shifted to the modeling phase.

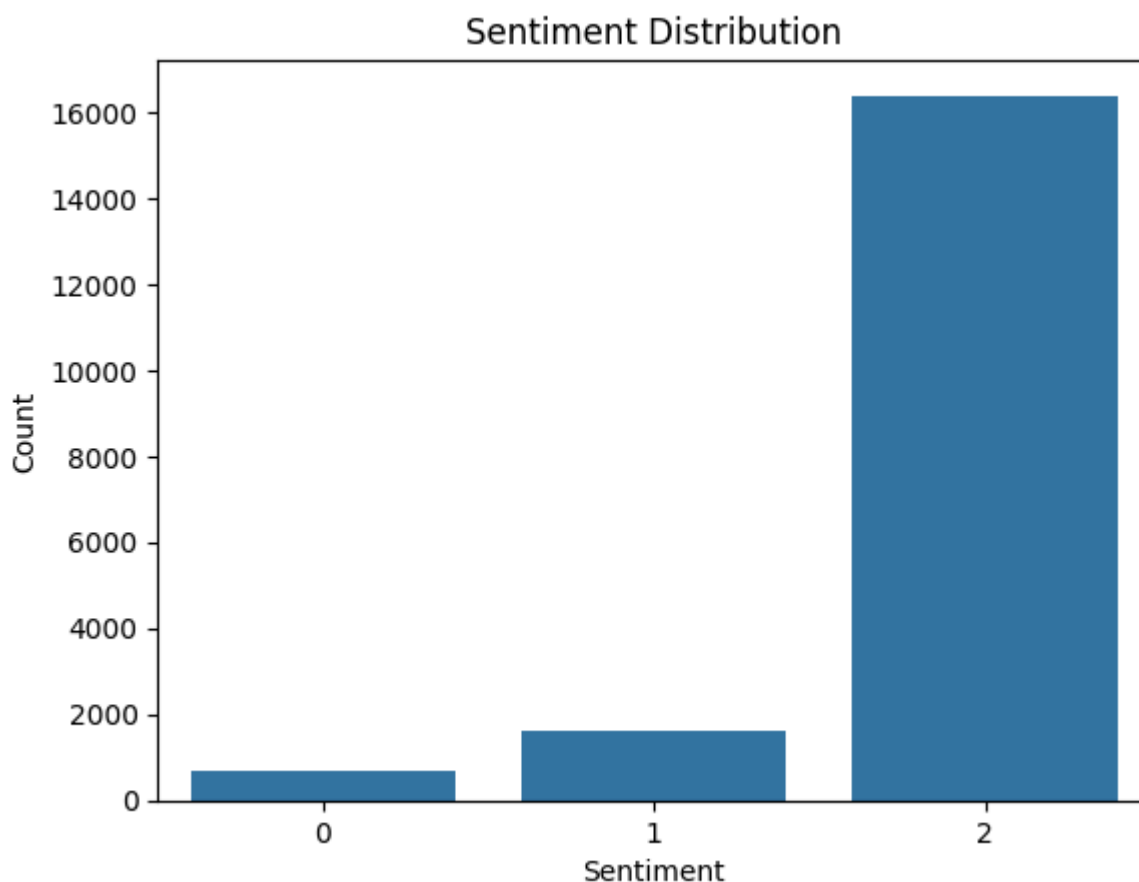
2.3 Data Preparation

The data was split into training(80%) and test(20%) sets using the Sci-kit learn library.

The text data which has undergone some preprocessing, now needs to be vectorized before we can use it for a model.

Different vectorization techniques like TF-IDF, Countvectorizer and pre-trained ones like word-to-vec were tried out with models like logistic regression and naive bayes classifier models and upon evaluating the results, it was decided to use Countvectorizer for the project.

Upon evaluating the count for each sentiment label, it was discovered that the data was heavily skewed towards positive sentiment.



To reduce the skewness, various sampling techniques like random oversampling, random undersampling and SMOTE were tried out with models like logistic regression and naive bayes classifier models and in the end , SMOTE was chosen.

2.4 Modeling and Tuning

After vectorizing the text data and applying sampling techniques to balance the dataset, we trained several classifiers to evaluate their performance.

The Logistic Regression Classifier gave an accuracy of 87%. The Multinomial Naive Bayes Classifier was also trained, but it did not perform as well as the Logistic Regression Classifier. Upon further tuning, the logistic Classifier gave 88% accuracy.

Advanced Classifiers were also tried. SVC was tested but did not provide the best results compared to other models. The RandomForest Classifier was trained with various parameters, providing strong performance but still not the highest accuracy. XGBoost outperformed other classifiers, achieving an accuracy of 93%. However, we discovered that the model was overfitted.

To leverage the strengths of different classifiers, an ensemble model was built using the VotingClassifier, which combines the predictions from multiple models:

- Logistic Regression Classifier
- RandomForest Classifier
- XGBoost Classifier

Grid Search was applied for fine-tuning the ensemble model, adjusting hyperparameters for each constituent model.

Logistic Regression Classifier: max iterations=1000.

C(Regularization Parameter)=0.1

Solver='liblinear'.

RandomForestClassifier: n_estimators(Number of trees in the forest)=100,

max_depth(Maximum depth of each tree)=10,

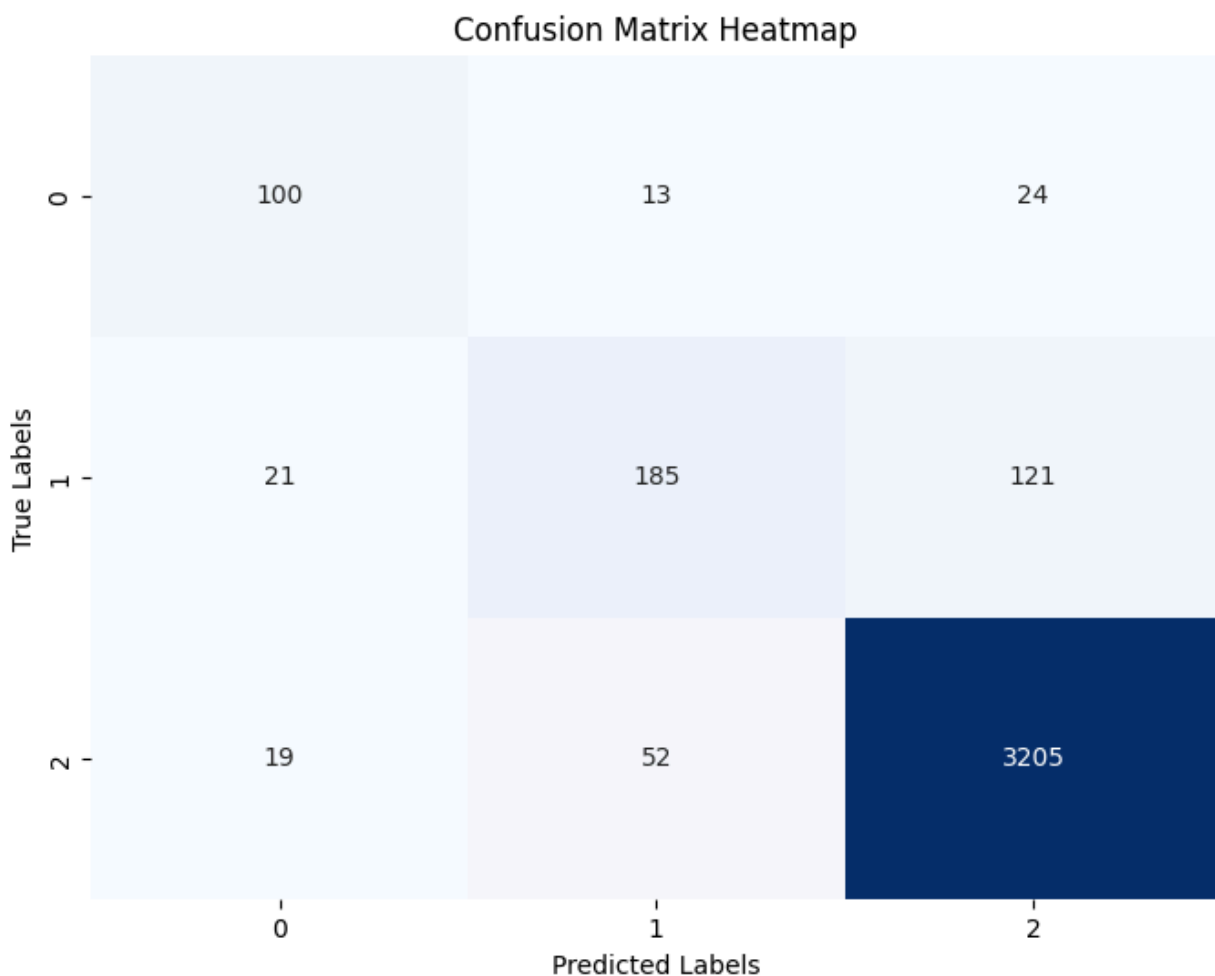
min_samples_split(Minimum number of samples required to split a node)=10,

min_samples_leaf(Minimum number of samples required at a leaf node)=4.

XGBClassifier: max_depth(Maximum depth of the tree)=4,
 reg_alpha(L1 regularization term)=0.1,
 reg_lambda(L2 regularization term)=0.1,
 n_estimators(Number of boosting rounds)=100.

Upon tuning, the accuracy was still 93% however, the overfitting issue was mostly resolved and the bias had decreased.

The model , however, even after resampling and tuning, still showed a bias towards the positive sentiment although the tuning managed to reduce it a little.



We decided to continue with this model for now and try advanced techniques in the future.

2.5 Web App

We decided to make just a basic sentiment classification app for now and not tailor it for a hotel database for now. To deploy the sentiment analysis model, we utilized Streamlit, a popular framework for building interactive web applications in Python. The deployment process involved the following steps:

- Development of Streamlit App:
 - Created a Streamlit app script that integrates with the trained sentiment analysis model.
 - Incorporated the necessary libraries such as streamlit, joblib for loading the model, and NLTK for text processing.
- Model Integration:
 - Loaded the pre-trained models using the joblib library.
 - Integrated the models into the Streamlit app to handle sentiment prediction based on user input.

The key features include:

User Input:

- Users can enter a review in the text input box.

Sentiment Prediction:

- Upon submission, the review is processed and classified by the sentiment analysis model.
- The app displays the predicted sentiment as "Positive," "Neutral," or "Negative."

Animated Emoji:

- Based on the predicted sentiment, an animated emoji is shown to visually represent the sentiment:
 - Positive Sentiment: 😊 (Smiling Face)
 - Neutral Sentiment: 😐 (Neutral Face)
 - Negative Sentiment: 😞 (Sad Face)

[Web-app link](#)

3. Literature Survey

3.1 Introduction to Sentiment Analysis

Sentiment analysis, also known as opinion mining, is a crucial task in natural language processing (NLP) that involves determining the sentiment expressed in text data. The sentiment could be positive, negative, or neutral. This technique is widely employed across industries such as marketing, customer service, and social media monitoring, where understanding public sentiment is vital for strategic decision-making.

Over time, various approaches have been developed for sentiment analysis. Early methods often relied on lexicon-based techniques, which use predefined lists of words associated with positive or negative sentiments to classify text. However, these methods lack the ability to capture context, making them less effective in understanding nuanced language.

In the travel and tourism industry, sentiment analysis has been applied to analyze customer reviews and feedback on hotels, airlines, and tourist attractions. Research has shown that customer sentiment is a strong predictor of business success, influencing factors such as booking rates and customer loyalty. This project leverages sentiment analysis to classify hotel reviews, providing insights that can be used to enhance service quality.

3.2 Technologies Used

Machine learning has introduced more robust models for sentiment analysis. Techniques such as Support Vector Machines (SVM), Naive Bayes, and Decision Trees have been extensively applied to sentiment classification tasks. These models require text to be transformed into numerical features, a process that is typically achieved using methods like Bag of Words (BoW) or Term Frequency-Inverse Document Frequency (TF-IDF).

CountVectorizer was used to convert the text data into a matrix of token counts. CountVectorizer is a simple and effective method that creates a vocabulary of all the words in the text and then encodes the text as a vector of word frequencies. This method captures the importance of words in the dataset but does not account for word order or semantics. It is widely used in traditional machine learning models as a feature extraction technique.

3.3 Challenges

One of the challenges in sentiment analysis is dealing with imbalanced datasets, where one sentiment class may dominate the others. This can lead to biased models that perform poorly on the minority class. To address this, we employed Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic samples for the minority class by interpolating between existing samples. This technique helps balance the dataset, leading to more robust and fair model performance.

Ensemble learning, which combines multiple models to improve overall performance, is another advanced technique used in sentiment analysis. In this project, we used a Voting Classifier ensemble that combines Logistic Regression, RandomForestClassifier, and XGBoost models. The Voting Classifier aggregates the predictions of these models to make a final decision, which often leads to improved accuracy and generalization compared to individual models. This approach is particularly effective when the base models capture different aspects of the data, thus complementing each other.

Despite significant advancements, sentiment analysis still faces challenges such as handling sarcasm, negations, and domain-specific language. Additionally, class imbalances, as addressed with SMOTE in this project, can complicate model training and evaluation. These challenges highlight the importance of selecting appropriate models and preprocessing techniques.

3.4 Conclusion

In this project, a combination of traditional and advanced machine learning techniques were utilized to classify customer reviews of hotels into positive, neutral, or negative sentiments. Text preprocessing, feature extraction using CountVectorizer were applied, and SMOTE was employed to address class imbalance. By using an ensemble of Logistic Regression, RandomForest, and XGBoost models, high accuracy while reducing overfitting and bias through careful hyperparameter tuning was achieved .

4. Result

After completing the preprocessing, modeling and app development phases of the project, the following results were achieved :

1. Ensemble Model:

- An ensemble model was built using a VotingClassifier that combined Logistic Regression, RandomForest, and XGBoost. This approach leveraged the strengths of each model, leading to a robust classifier.
- Final Ensemble Model Performance:
 - Accuracy: 93%
 - Precision, Recall, and F1-Score: These metrics also improved, with minimal overfitting observed. The final model effectively balanced bias and variance and resolved overfitting.

2. Impact of Feature Engineering:

- Text Vectorization with CountVectorizer: The features derived from the text data played a crucial role in the model's performance. By converting text into a sparse matrix of token counts, the model could capture the frequency and presence of words that are strong indicators of sentiment.
- SMOTE for Imbalanced Data: Implementing SMOTE was key in improving the model's ability to correctly classify minority classes, particularly the negative and neutral sentiments, which were underrepresented in the dataset.

3. Model Deployment:

- **Web Application:** The final model was deployed using Streamlit, enabling users to input a review and receive a sentiment prediction.

The combination of effective text preprocessing, feature engineering, and advanced modeling techniques allowed us to build a high-performing sentiment analysis model. The final model not only achieved high accuracy but also demonstrated robustness in handling imbalanced data and complex sentiment patterns. However, there still exists a bias in the model, which needs to be resolved in the future.

5. Conclusion

This sentiment analysis project effectively applied NLP techniques and machine learning models to classify customer reviews into positive, neutral, or negative sentiments. Key outcomes include:

1. **Preprocessing:** Essential text processing steps, such as tokenization and lemmatization, transformed raw data into useful features using CountVectorizer.
2. **Handling Imbalance:** SMOTE was utilized to address class imbalance, improving model accuracy and ensuring fair representation of all sentiment classes.
3. **Model Performance:** XGBoost achieved the highest accuracy at 93%. Combining models with VotingClassifier further enhanced performance and reduced overfitting.
4. **Deployment:** The model was deployed via Streamlit, providing an interactive interface for users to input reviews and receive sentiment predictions with visual emojis.

Overall, the project demonstrated the effectiveness of integrating NLP and machine learning techniques to create a robust sentiment analysis tool, offering valuable insights into customer feedback and improving service quality.

6. Future Work

We will try better feature engineering methods, advanced vectorization techniques and deep learning algorithms for modeling in the future to reduce bias and improve semantic understanding and then improve the web-app to tailor it to make reports based on a hotel database so that the management can derive insights.

7.References

1.NumPy Documentation:

<https://numpy.org/doc/stable/>

2.pandas Documentation:

<https://pandas.pydata.org/pandas-docs/stable/>

3.Matplotlib Documentation:

<https://matplotlib.org/stable/contents.html>

4.CountVectorizer:

https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.CountVectorizer.html#countvectorizer

5.SMOTE (Synthetic Minority Over-sampling Technique):

https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.SMOTE.html

6.VotingClassifier:

<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.VotingClassifier.html>

7.XGBoost:

<https://xgboost.readthedocs.io/en/latest/>